SKANDA SHREESHA PRASAD

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OBJECTIVE

A research-oriented student aiming to contribute to AI Research. Currently searching for research-based learning experiences.

EDUCATION

- **New York University – Courant Institute for Mathematical Science,** New York City, USA. July 2025 – Present

Master of Science in **Computer Science**
Related Courses: Programming Languages; Operating Systems; Mathematical Foundations of Deep Learning and Large Language Models (Auditing); Bayesian Machine Learning;

- **PES University – Ring Road Campus,** Bengaluru, India. Sep 2021 – May 2025

Bachelor of Technology in **Computer Science and Engineering**.
Related Courses: Statistics for Data Science; Automata Formal Languages, and Logic; Linear Algebra; Data Analytics; Big Data; Machine Intelligence; Cloud Computing; Topics in Deep Learning; Natural Language Processing; Large Language Models;
Special Topic Courses: CIE L1: Getting Started with Entrepreneurship; CIE L2: Building a Lean Startup; TWSD: Telling Stories with Data.

EXPERIENCE

Bosch Global Software Technologies Limited (BGSW), Bengaluru
AI Research Intern March 2025 – July 2025
• AI Research under a Centre of Excellence in Bosch
• Worked on the project, "Weather Resilient ADAS: Improving Perception in Rain, Fog and Low Visibility"
• Submitted three invention reports, all due for patenting.
Generative AI Intern June 2024 – August 2024
• Built a Generative AI powered chatbot for an upcoming Data Accelerator system being launched by BGSW.
• Involved configuring metadata, managing metadata, and retrieving information through the chatbot.
Machine Learning and Computer Vision Intern June 2023 – July 2023
• Worked on Optimizing Production Line by Eliminating Testing Stations using Data from Test Stations.
• Built Computer Vision Model to differentiate between Microscopic Parts needed for different diesel engines. To be implemented in the production line soon.

Department of Computer Science and Engineering, PES University, Bengaluru
Teaching Assistant July 2024 – May 2025
• Teaching Assistant for a newly introduced 4 credit course, Generative AI and its Applications and Mathematics for Computer Science Engineers (Previously known as Statistics for Data Science).
• Involved in making slide decks, preparing coding assignments and guiding students in their course projects, preparing and evaluating assignments on Optimization in Statistics.

RESEARCH PROJECTS

Enhancing Multi-Agent LLM Frameworks: A Context-Aware Approach for Securing Dynamic Agent Collaboration
Jan 2024 – Dec 2024
PES University, Bengaluru
Guide: Prof. Dr S Natarajan

- Capstone Project on building an LLM Agents framework which are context aware and are robust enough to prevent jailbreak.
- Paper published in ICAART-2025.

Code-Mixed LLM for Natural Language Generation and Translation Oct 2023 – Jan 2025
PES University, Bengaluru
Guide: Prof. Dr Uma D

- A University-funded project on building a custom LLM for generating code-mixed Indian languages with English.
- Completed building English to Hindi-English codemixed translator.
- Currently collecting/synthetically generating datasets of English-Kannada code-mixing, wherein Kannada is a low-resource language.

RoboCrop Jan 2023 – Jul 2023
PES University, Bengaluru
Guide: Prof. Prasad B Honnavalli, Prof. Revathi G P

- A Precision Agriculture CNC Farming Robot based on FarmBot.io and the white paper published by Rory Aronson.
- Currently being used for building 'YODA – Your Own Data' for Computer Vision Based Projects involving plant datasets.

Handwritten Kannada Text to Typed Text Conversion Jul 2023 – Mar 2024

PES University, Bengaluru

- A full-fledged Kannada OCR model.
- Worked on a custom dataset of more than 75 thousand images.

SELECT INTERNSHIP PROJECTS

Weather Resilient ADAS: Improving Perception in Rain, Fog and Low Visibility

Mar 2025 – Jul 2025

Bosch Global Software Technologies Private Limited, Bengaluru

Guide: Dr. Anil Kumar A

- Worked on building Computer Vision systems that are much more efficient in unfavorable lighting conditions for ADAS.
- Submitted three invention reports in four months that are currently being patented by the company.

DataNexusBot: A Gen AI Powered Intelligent Metadata Management System for Enterprise Data Platform

Jul 2024 – Aug 2024

Bosch Global Software Technologies Private Limited, Bengaluru

Guide: Mr. Philip Ronaldo Dang, Mr. Harsha Puttur Rajagopal

- LLM-Powered Chatbot Interface for interacting with the DataNexus system, an upcoming Data Accelerator by BGSW for streamlined configuration of the entire dataflow and data pipeline.

Differentiation between Ground and Turned Finishing of a Miniscule Mechanical Part June 2023 – Jul 2023

Bosch Limited, Bengaluru

Guide: Mr. Kumar S

- Automation of the differentiation between Ground and Turned Finishing of a Miniscule Mechanical Part in a Diesel Pump that had to be done manually using a High-Power Microscope. Produced results of about 96% accuracy.

SELECT COURSE PROJECTS

Context-Aware LLM for Retrieving Medical Literature to Assist Diagnosis Feb 2024 – May 2024

PES University, Bengaluru

- Working on making LLMs pre-trained on PubMed data Context-Aware using RAG architecture.
- Curated a corpus of medical textbooks to retrieve information.

Detection of Hallucinations in LLMs Feb 2024 – May 2024

PES University, Bengaluru

- Working on detecting hallucinations in open-source LLMs to detect hallucinations.
- Uses Step-Back Question Answering Prompting technique to generate output.
- Compares output generated with an evaluation dataset to detect hallucinations.

Yet Another Distributed File System Sep 2023 – Nov 2023

PES University, Bengaluru

- Built a custom Distributed File System with three datanodes.
- Contained a namenode, datanodes, metadata, etc.
- File System deployed on Docker Containers.

TECHNICAL SKILLS AND CERTIFICATIONS

- **Hardware Design Languages:** Icarus Verilog, ARM7.
- **Programming Languages:** Python, R, C, Java, SQL, MATLAB, JavaScript, CSS, HTML, Prolog. **Database:** MongoDB, MySQL.
- **Tools:** MATLAB, Tableau, Jupyter, Colab, Docker, Kubernetes, AWS, RAFT, Git, GitHub.
- **Softwares Packages:** OpenCV, MATLAB, NLTK, Numpy, Pandas, Sci-Kit Learn, TensorFlow, PyTorch, etc.
- **Certificates:** Introduction to Machine Learning - Numpy and Pandas; Supervised Machine Learning: Regression and Classification; Data Structures: Problem Solving (Intermediate); A Beginner's Guide to Linux Kernel Development (LFD103). Currently learning Generative AI and Reinforcement Learning. Completed 6 Harvard ManageMentor courses by Harvard Business Review.
- **Micro-Controller boards:** Arduino Uno & Mega, Raspberry-Pi 3 & 4 & Pico.

LEADERSHIP

Team Leader of 7 in the RoboCrop project, managing three different domains from different branches of engineering, namely, Mechanical, Electronics, and Computer Science.

Campus Director of PES University's Millennium Fellowship Cohort.

- Campus Director of PES University's Cohort for the Millennium Fellowship Program.
- Worked on social projects involving education using technology.

Part of the Reinforcement Learning and LLM Special Interest Groups of PES University's CDSAML Lab.

- Actively participating in the discussions and meetings of the respective SIGs.
- Actively peer mentor and advise Freshman and Sophomore year students.